

Intro to Git & GitHub

Week 1 - 06/15/2026

LEARNING GOALS:

Understand the “Reproducibility Crisis”
Learn the difference between Git & GitHub
Familiarize yourself with essential vocabulary

How do you know if work is trustworthy?

A Case Study: *Jonathan Pruitt*



University investigation found prominent spider biologist fabricated, falsified data

Co-authors of now-retracted papers from Jonathan Pruitt were sent summarized findings that the scientist violated McMaster University's research integrity policy

11 MAY 2023 • 7:20 PM ET • BY [CLAUDIA LÓPEZ LLOREDA](#)

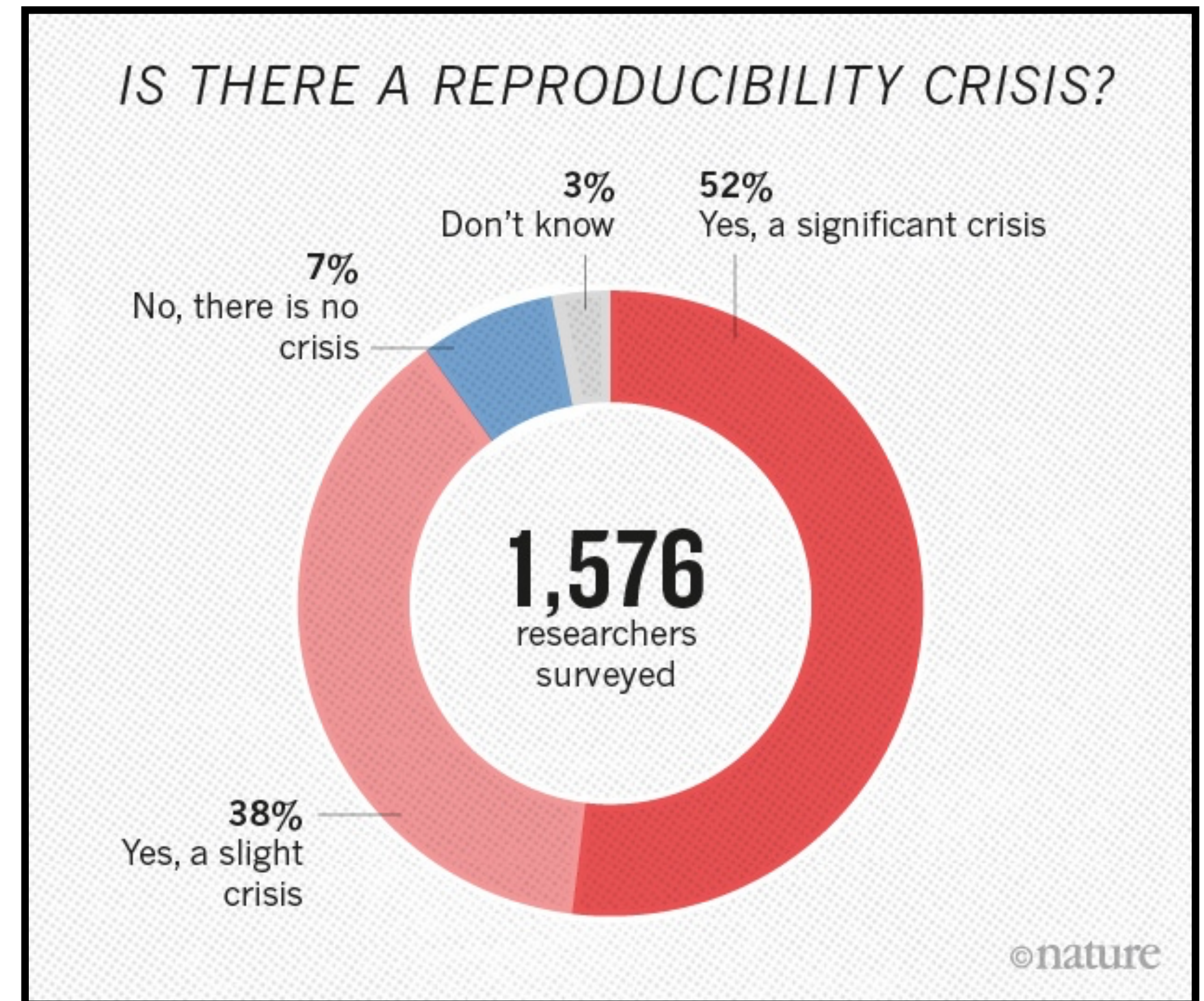
Behavioral ecologist Jonathan Pruitt's PhD dissertation withdrawn

Fraud is hard to catch since scientists **tend to trust** one another

But *really*, how common are mistakes?

The Reproducibility Crisis

A phenomena in scientific fields where published results cannot be reliably replicated. Most associated with medical research, but shows up across many fields.



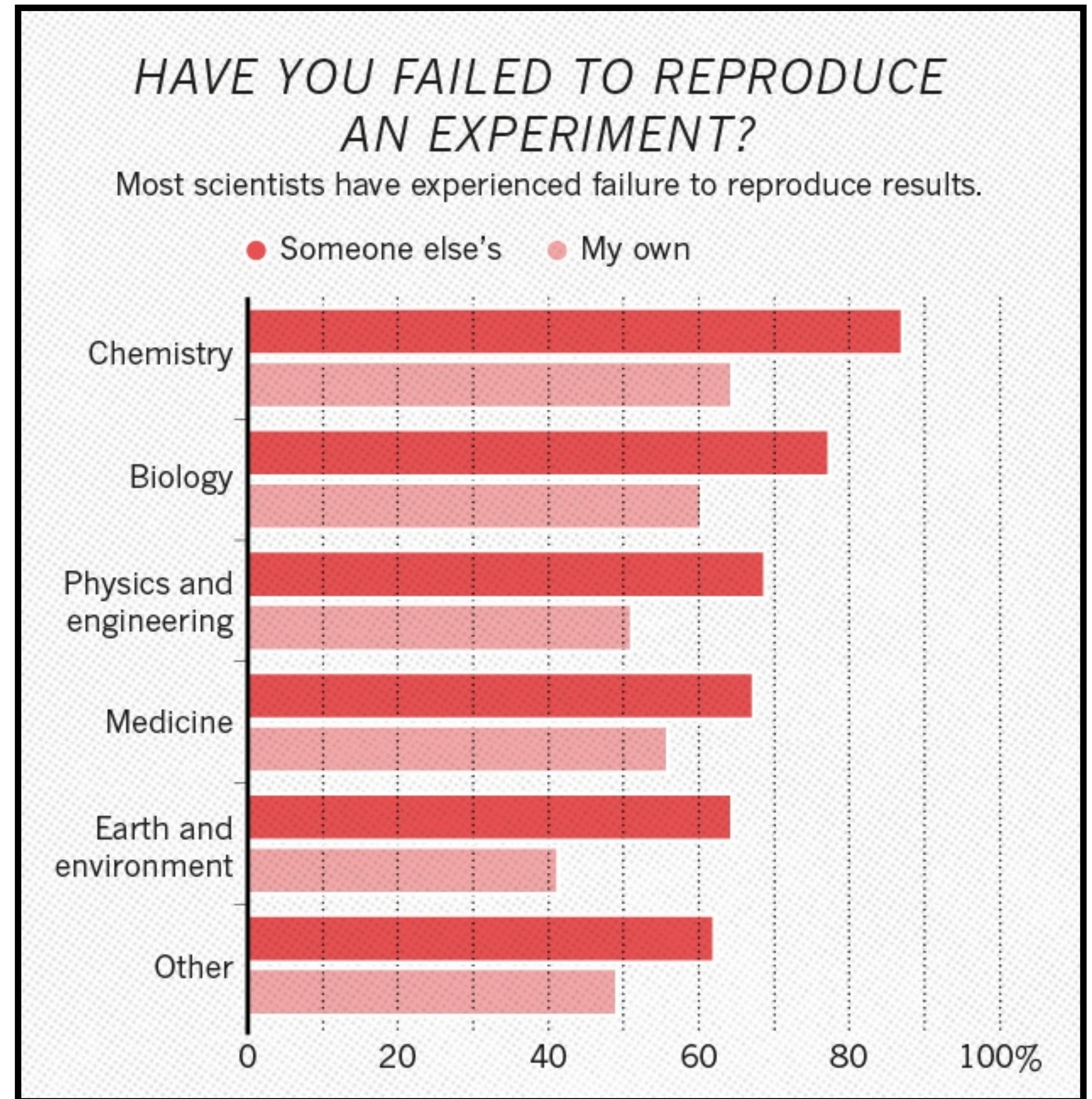
from Monya Baker; "1,500 scientists lift the lid on reproducibility"

The Reproducibility Crisis

A phenomena in scientific fields where published results cannot be reliably replicated. Most associated with medical research, but shows up across many fields.

Across many disciplines, scientists sometimes ***fail*** to ***replicate results***.

This includes their own experiments.



from Monya Baker; "1,500 scientists lift the lid on reproducibility"

How do I establish trust?

How Do You Write Robust, Reproducible Code?

Ensuring all code follows a “style guide”

Unit / Functional Testing

Providing comments + docstrings

Explicitly handling errors / exceptions

Writing small single-purpose functions

⋮

*Each of these topics will
be discussed this Summer!*

A Very Basic Overview of Git / GitHub



is not the same as



- version tracking software
- runs entirely on your device
- open-source project

- file hosting service
- runs entirely on remote servers
- owned by Microsoft

Your computer uses...



...to track changes

You can use...



...to easily save this info



Two Types of Repositories



Local Repository



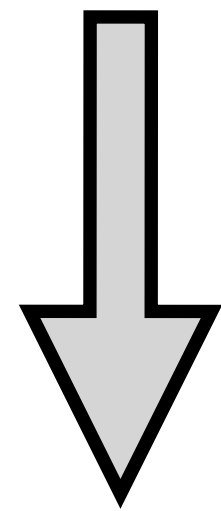
Remote Repository

TrackedFolder

- code.py
- README.md
- tests.sh

GitHub

(remote repository)



```
> git clone repository-name
```

TrackedFolder

- code.py
- README.md
- tests.sh

Your Computer

(local repository)

Setting up a local repository

If a repository already exists, then you need to ***clone*** the ***remote repository***.

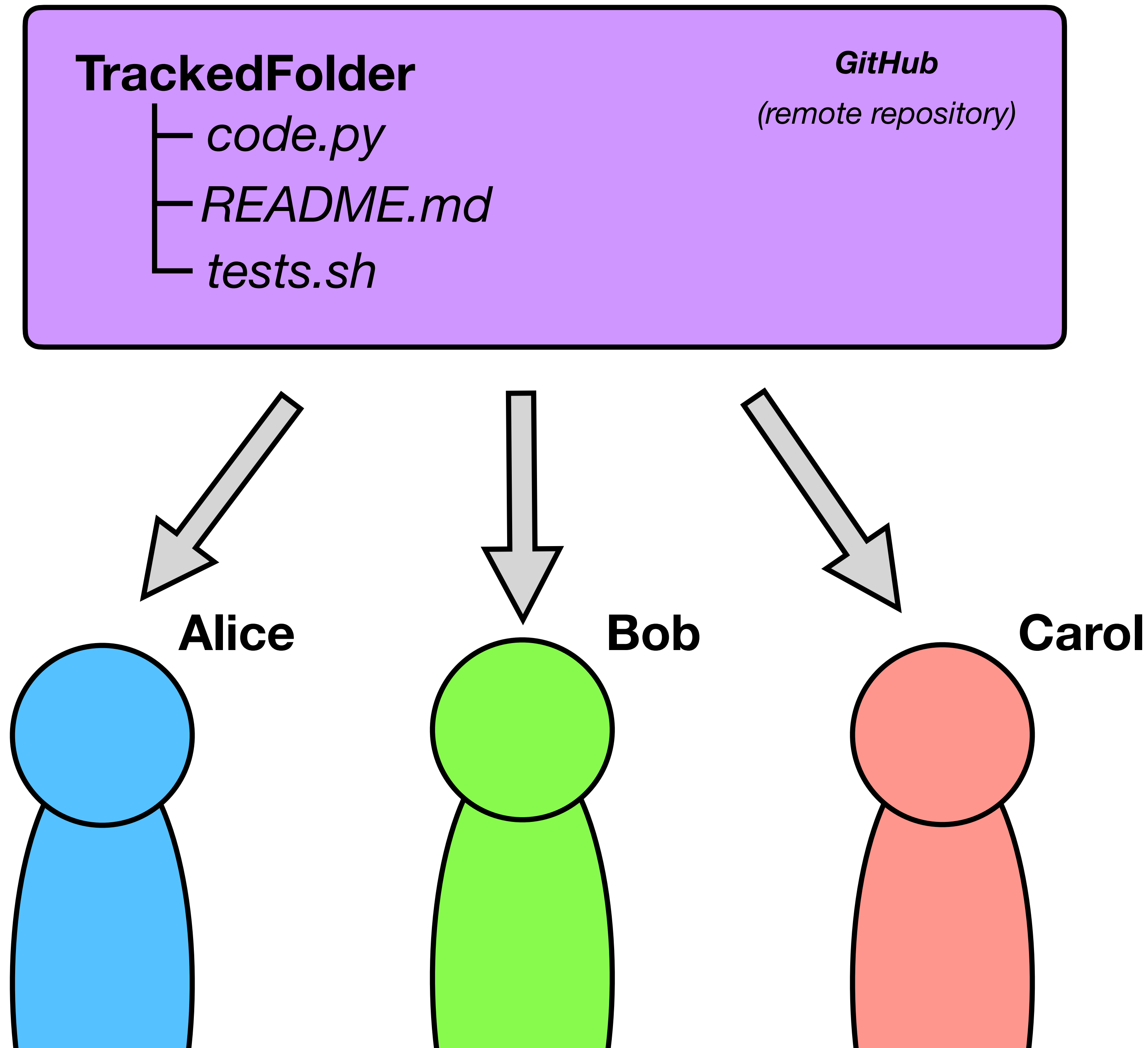
This command will create a tracked folder in your terminal's current directory.

Once cloned, you should not move the folder! This will break Git's ability to track files.

Can multiple people clone?

We will create three imaginary people to help explain how Git & GitHub work.

Since Alice, Bob, and Carol all cloned the remote repository, they all have the same files on their own devices.



TrackedFolder

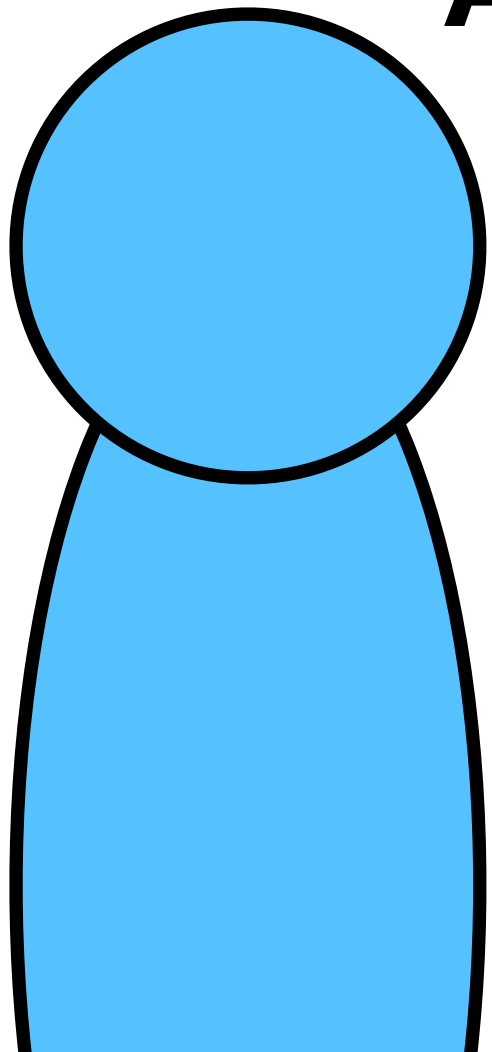
- *code.py*
- *README.md*
- *tests.sh*
- **data.csv**

(untracked change)

Alice's Computer

(local repository)

Alice



How do I make changes?

Let's say Alice added a file called "data.csv" to her local repository. At this point, Git has no idea what you have done!

TrackedFolder

- code.py
- README.md
- tests.sh
- data.csv (staged change)

Alice's Computer
(local repository)

Alice

```
> git add data.csv
```

How do I make changes?

Let's say Alice added a file called "data.csv" to her local repository. At this point, Git has no idea what you have done!

To tell Git that you've made a change, you need to ***add*** the file. This prepares Git for making a "save state," but has not done so yet.

TrackedFolder

- code.py
- README.md
- tests.sh
- data.csv (committed change)

Alice's Computer

(local repository)

Alice

```
> git add data.csv
```

```
> git commit -m "Added data file!"
```

How do I make changes?

Let's say Alice added a file called "data.csv" to her local repository. At this point, Git has no idea what you have done!

To tell Git that you've made a change, you need to **add** the file. This prepares Git for making a "save state," but has not done so yet.

After adding all changed files, you need to **commit** it to finalize the change.

TrackedFolder

- *code.py*
- *README.md*
- *tests.sh*
- *data.csv*

GitHub

(remote repository)

Alice

```
> git add data.csv
```

```
> git commit -m "Added data file!"
```

```
> git push origin main
```

Updating the remote repo.

Once you finished all your changes, you can update the **remote repository** by **pushing** your local version.

This can take a couple seconds to show up on GitHub, but is usually very fast.

TrackedFolder

- code.py
- README.md
- tests.sh
- data.csv

GitHub

(remote repository)

A discrepancy in repositories...

Remember, Bob and Carol only have the version of the code that existed before Alice made her changes!

If they were to try and ***push*** right now, they would get a bunch of errors and warnings.

TrackedFolder

- code.py
- README.md
- tests.sh

Bob / Carol Computer
(local repository)

Bob

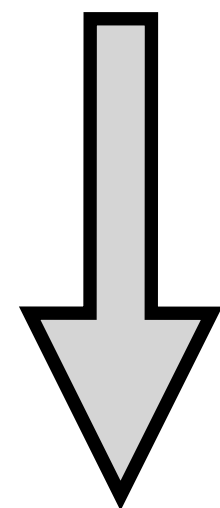
Carol

TrackedFolder

- code.py
- README.md
- tests.sh
- data.csv

GitHub

(remote repository)



Bob

```
> git pull origin main
```

```
> git push origin main
```

Updating your local repository

Remember, Bob and Carol only have the version of the code that existed before Alice made her changes!

If they were to try and ***push*** right now, they would get a bunch of errors and warnings.

For relatively simple conflicts, you can fix this discrepancy by ***pulling*** the remote repository.

Trying to Handle Conflicts

Let's say Carol and Bob both made changes to overlapping sections of *code.py*. It isn't easy to know which changes to keep...

If you use VS Code, you will be asked to select which changes to keep for every detected conflict.

In the future, we will talk about how using GitHub can prevent thorny conflicts from arising!

TrackedFolder

- *code.py* (Bob's version)
- *README.md*
- *tests.sh*
- *data.csv*

GitHub

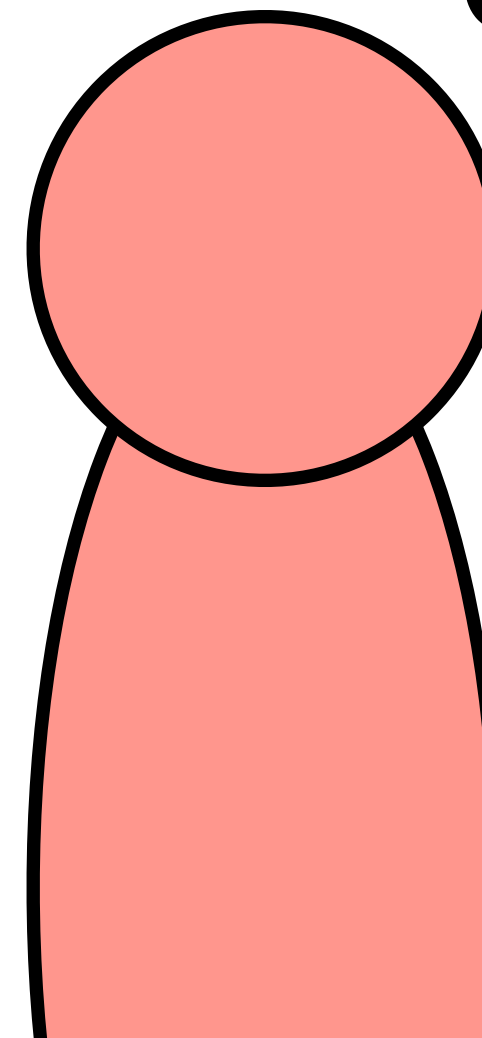
(remote repository)

TrackedFolder

- *code.py* (Carol's version)
- *README.md*
- *tests.sh*

Carol Computer
(local repository)

Carol



Continuous Integration Rule:

Merge your changes **frequently** so that file conflicts are small

Getting Started: *Let's Clone a Repository!*

Autumn10677 / practice_collaborative_coding

Type / to search

<> Code

Issues

Pull requests

Agents

Actions

Projects

Wiki

Security and quality

Insights

Settings

practice_collaborative_coding

Public

Pin

Watch 0

Fork 0

Star 0

main

1 Branch

0 Tags

Go to file

Add file

<> Code

Autumn10677

Updated README

a308c58 · 2 weeks ago

2 Commits

README.md

Updated README

2 weeks ago

README

Collaborative Coding Practice

Brief Overview

We will be using this GitHub repository to learn about and practice collaborative coding and continuous integration!

About

No description, website, or topics provided.

Readme

Activity

0 stars

0 watching

0 forks

Releases

No releases published

Create a new release

Getting Started: *Let's Clone a Repository!*

Autumn10677 / practice_collaborative_coding

Type / to search

+

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Security and quality

Insights

Settings

practice_collaborative_coding

Public

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Watch 0

Fork 0

Star 0

main

1 Branch

0 Tags

Go to file

Add file

Code

Autumn10677 Updated README

README.md Updated README

README

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No releases published

Create a new release

Local

Codespaces

Clone

HTTPS SSH GitHub CLI

https://github.com/Autumn10677/practice_ci

Clone using the web URL.

Open with GitHub Desktop

Download ZIP

Getting Started: *Let's Clone a Repository!*

```
> git clone https://github.com/Autumn10677/practice_collaborative_coding.git
```

The screenshot shows the GitHub interface for the repository `practice_collaborative_coding` by user `Autumn10677`. The repository is public and has 1 branch, 0 tags, and 0 stars. The `Code` button is highlighted in green. A dropdown menu is open, showing the `Clone` option selected. The `HTTPS` option is selected, and the URL `https://github.com/Autumn10677/practice_collaborative_coding.git` is highlighted with a red box. The repository has a README file titled `Collaborative Coding Practice`. The `Code` button is also visible in the top right corner of the repository page.

Code Issues Pull requests Agents Actions Projects Wiki Security and quality Insights Settings

`practice_collaborative_coding` Public Pin Watch 0 Fork 0 Star 0

main 1 Branch 0 Tags Go to file Add file Code

Autumn10677 Updated README

README.md Updated README

README

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Readme Activity 0 stars 0 watching 0 forks

Releases

No releases published [Create a new release](#)

Getting Started: *Updating the README.md*

README.md

README.md > abc

Collaborative Coding Practice

> abc

Getting Started

1

Collaborative Coding Practice

2

3

List of Contributors:

4

5

Brief Overview

6

We will be using this GitHub repository to learn about and practice collaborative coding and continuous integration!

7

Collaborative Coding Practice

List of Contributors:

Brief Overview

We will be using this GitHub repository to learn about and practice collaborative coding and continuous integration!

Getting Started: *Updating the README.md*

```
README.md ×  
README.md > abc # Collaborative Coding Practice > abc ### Getting Started  
1 # Collaborative Coding Practice  
2  
3 ### List of Contributors: ← Add your name here!  
4  
5 ### Brief Overview  
6 We will be using this GitHub repository to learn about and practice collaborative coding  
7 and continuous integration!
```

Getting Started: *Updating the README.md*

```
README.md ×
README.md > abc # Collaborative Coding Practice > abc ### Getting Started
1 # Collaborative Coding Practice
2
3 ### List of Contributors:
4
5 ### Brief Overview
6 We will be using this GitHub repository to learn about and practice collaborative coding
7 and continuous integration!
```



Add your name here!

```
> git add README.md
```

```
> git commit -m "Added '___' to README"
```

```
> git push origin main
```

Save and upload your changes!

A Handy Cheat Sheet

git...

...clone First-time **download and setup** of local a repository

...add **Stages** a changed file for saving

...commit Create a **new save state** with all new changes

...push **Update remote** repository **using local** changes

...pull **Update local** repository **using remote** changes

Next Time...

Introduction to Branches & Style Guides